HyperNiche: Learning Spatially Localized Higher-Order Cellular Neighborhoods*

Md Ishtyaq Mahmud[1], Anish Turumala Venkata[2], Kumaraswamy Naidu Chitrala[1,3], Tania Banerjee[1,4]

1 Department of Electrical and Computer Engineering, University of Houston, Texas 77004, USA
2 Department of Computer and Information Science, University of Florida, Florida 32611, USA
3 Department of Engineering Technology, University of Houston, Texas 77479, USA
4 Department of Information Science Technology, University of Houston, Texas 77479, USA
E-mail: mmahmud4@cougarnet.uh.edu, anishve9@gmail.com, kchitral@central.uh.edu, tbanerjee@uh.edu

Spatial transcriptomics enables the study of cellular niches: localized multicellular environments that may contain molecularly distinct cell populations. Existing spatial graph methods represent these environments through pairwise relationships, whereas many hypergraph methods construct fixed hyperedges using spatial or molecular similarity. Such constructions cannot adapt their incidence structure during training and may favor homogeneous groupings. We introduce HyperNiche, an end-to-end hypergraph neural network that jointly learns cellular representations and spatially constrained soft hyperedges. Each cell anchors a candidate hyperedge, and an asymmetric compatibility function assigns memberships among its spatial neighbors without requiring the anchor and member cells to be transcriptionally similar. Adaptive hyperedge weighting, hypergraph message passing, and structural regularization promote selective and reproducible memberships. We evaluate HyperNiche on Xenium breast- and lung-cancer cohorts containing 42 tissue cores from nine patients and 84,177 cells after quality control. Under patient-level cross-validation, HyperNiche achieves the highest agreement with independently established cell-type annotations among the evaluated graph and hypergraph methods. The learned niches are spatially compact, compositionally selective, and reproducible across random initializations, although enrichment beyond matched spatial neighborhoods varies between cohorts. Ablations show that learned asymmetric memberships provide the largest predictive benefit, whereas message passing and degeneracy regularization are important for producing selective and reproducible niches. These results support HyperNiche as a framework for learning localized higher-order cellular organization without imposing cell-type homogeneity. Code will be available at `https://github.com/ishtyaqmahmud/HyperNiche`.

Keywords: Spatial transcriptomics; cellular niches; hypergraph neural networks; higher-order cellular interactions; tumor microenvironment; learned hypergraph structure

*This project was supported by the National Center for Advancing Translational Sciences (NCATS), National Institutes of Health, through Grant Award Number UM1TR004539. The content is solely the responsibility of the authors and does not necessarily represent the official views of the NIH.

1. Introduction

Spatial transcriptomics enables gene expression to be studied while preserving the physical organization of cells within tissue. This spatial context is essential for investigating cellular niches: localized microenvironments in which multiple cell populations coexist and potentially coordinate biological functions. Such niches are particularly important in tumors, where malignant, immune, stromal, and vascular cells can occupy shared microenvironments while retaining distinct molecular identities. Consequently, a cellular niche is not necessarily a homogeneous cluster of transcriptionally similar cells; it may instead be a spatially localized, compositionally selective group containing cells with complementary roles. Elucidating this higher-order organization is important for understanding tissue architecture and the tumor microenvironment.[1,2]

Many computational approaches represent spatial transcriptomics data using pairwise graphs, commonly constructed from spatially adjacent cells, and apply graph neural networks (GNNs) to learn cellular representations.[3,4] Although these methods effectively incorporate local spatial information, pairwise graphs express a multicellular structure only as a collection of binary relationships.[5,6] Moreover, conventional graph construction and message passing often favor *homophily*, under which neighboring cells are expected to have similar features or representations.[7,8] This assumption is well suited to identifying homogeneous spatial domains but is restrictive for cellular niches composed of different cell types. Similarity-driven aggregation can suppress, rather than preserve, the heterogeneity that defines these multicellular microenvironments.[9,10]

Hypergraphs provide a natural representation for such organization because a single hyperedge can connect multiple cells simultaneously.[11–13] Hypergraph neural networks (HGNNs) can therefore model higher-order relationships that are not represented explicitly by ordinary graphs.[14] Nevertheless, existing spatial hypergraph approaches commonly construct their hyperedges before model training using fixed procedures such as spatial nearest neighbors, feature similarity, densest-subgraph extraction, or deterministic hyperedge decomposition.[2,15] These predefined incidence structures cannot adapt to the downstream learning objective. When their construction is driven primarily by similarity, they also inherit a homophilic bias and may preferentially group cells that already resemble one another.

We introduce *HyperNiche*, a hypergraph neural network that jointly learns cellular representations and spatially constrained hypergraph structure. Each cell serves as the anchor of a candidate hyperedge whose potential members are restricted to a local spatial neighborhood. Within this neighborhood, HyperNiche learns soft memberships through an asymmetric anchor–member compatibility function that combines molecular representations with relative spatial information. This compatibility formulation does not require an anchor and its members to be similar, allowing distinct cell types to participate selectively in the same learned niche. Adaptive hyperedge weights and hypergraph message passing then propagate information through the inferred higher-order structure. Structural regularization discourages trivial, overly dense, or degenerate memberships and promotes localized, interpretable hyperedges.

A central challenge in evaluating learned cellular niches is that agreement with reference cell-type labels and biological niche validity are related but distinct objectives. Cell-type labels describe individual cellular identities, whereas a niche can legitimately contain several cell types. Accordingly, we evaluate HyperNiche at two complementary levels. First, we assess whether its learned representations recover independently established cell-type annotations on held-out patients. Second, we characterize the inferred niches through cross-seed reproducibility, spatial extent, composition, and local enrichment. These structural measurements are interpreted as evidence of biological plausibility rather than proof that every inferred hyperedge represents a distinct biological program.

We evaluate HyperNiche on Xenium spatial transcriptomics cohorts from non-small cell lung cancer and breast cancer, comprising 42 tissue-microarray cores from nine patients and more than 84,000 cells after quality control. To prevent patient-level information leakage, all models are assessed using leave-one-patient-out cross-validation, with every tissue core from the same patient retained in a single partition. Under a common downstream clustering protocol, HyperNiche achieves the highest agreement with the reference annotations in both cohorts among the evaluated spatial graph and hypergraph methods. Its learned niches are spatially localized and reproducible across random initializations, while exhibiting selective cellular compositions and local enrichment.

Component ablations further distinguish predictive accuracy from meaningful niche formation. Replacing learned memberships with static spatial neighborhoods causes the largest predictive degradation, while symmetric similarity produces less accurate and less compositionally selective representations. Removing message passing or degeneracy regularization can preserve cell-type prediction while producing nonselective or irreproducible niches, demonstrating that predictive performance alone is insufficient for evaluating niche-learning methods. Sensitivity analyses additionally show that the principal predictive conclusions remain stable across reasonable neighborhood sizes and regularization settings, although niche reproducibility is more sensitive to the structural regularization strength.

The principal contributions of this work are as follows:

(1) We formulate cellular-niche discovery as learning spatially localized higher-order cell relationships that may span multiple cell types, rather than detecting compositionally homogeneous spatial domains.
(2) We develop HyperNiche, an end-to-end differentiable framework that learns anchor-centered hyperedge memberships through asymmetric compatibility rather than fixed similarity-based construction.
(3) We introduce adaptive hyperedge weighting, spatially constrained membership learning, and structural regularization to support selective and reproducible niche formation.
(4) We establish a leakage-controlled, leave-one-patient-out evaluation across two cancer spatial-transcriptomics cohorts, complemented by component ablations and parameter-sensitivity analyses.
(5) We evaluate both annotation recovery and the structural properties of the learned niches, showing that strong prediction and biologically plausible niche formation are complementary but non-equivalent outcomes.

2. Related Work

Graph Representation Learning for Spatial Omics. Spatial transcriptomics has significantly advanced our understanding of tissue microenvironments by enabling spatially resolved measurement of gene expression.[1,2] Many computational approaches rely on Graph Neural Networks (GNNs), such as STAGATE[3] and DeepST,[4] which construct spatial neighborhood graphs based on physical adjacency. While effective for modeling local structure, these methods are inherently limited by their pairwise formulation. In particular, they cannot directly represent higher-order relationships among multiple cells and typically rely on similarity-based message passing, which promotes homophily. However, cellular niches may be heterophilic, comprising distinct cell types with complementary functional roles within shared microenvironments.

Hypergraph Neural Networks for Higher-Order Modeling. To address these limitations, Hypergraph Neural Networks (HGNNs) extend graph-based models by allowing hyperedges to connect multiple nodes simultaneously.[14] Recent methods leverage this framework to model multi-cellular interactions in spatial omics. For example,[15] constructs hyperedges from top-K densest overlapping subgraphs defined by spatial and histological similarity, and combines HGNN message passing with denoising autoencoders to learn robust embeddings. Similarly, HyperSTAR[2] refines hypergraph topology through gene expression-guided hyperedge decomposition, improving boundary delineation and spatial domain identification while still relying on heuristic hyperedge construction. While these approaches enhance expressivity by modeling higher-order relationships, they still rely on predefined structural heuristics to construct hyperedges.

Static Hypergraph Construction and Its Limitations. A common limitation across existing HGNN-based methods is the reliance on static, non-differentiable procedures—such as k-nearest neighbors, densest subgraph extraction, or deterministic decomposition—to define hyperedges prior to training.[14] These heuristics fix the hypergraph topology before optimization, limiting the model's ability to adapt hypergraph topology during learning.[16] Moreover, because hyperedge construction is often driven by feature similarity or spatial proximity, these methods implicitly inherit homophilic inductive biases, limiting their ability to recover heterogeneous cellular niches.

Parametric and Geometric Hypergraph Models. Beyond application-driven approaches, recent theoretical work has explored parametric spatial hypergraph models that interpolate between pairwise graphs and higher-order structures.[17] These models generate hyperedges by clustering local neighborhoods under a spatial cohesion parameter, enabling controlled analysis of higher-order effects on processes such as diffusion and epidemic spreading. However, hyperedge formation in these frameworks remains governed by deterministic clustering algorithms, resulting in fixed incidence structures that are not optimized end-to-end.

Spectral and Diffusion-Based Hypergraph Representations. Another line of work focuses on learning multiscale representations over hypergraphs using spectral and diffusion-based techniques. For instance,[18] introduces hypergraph diffusion wavelets for learning multi-scale structural properties of cellular niches. By constructing hyperedges from k-hop spatial neighborhoods and applying random-walk-based diffusion operators, this approach extracts rich geometric features for downstream tasks. Despite their representational power, these methods still operate on heuristically defined, static hypergraph structures and do not address the problem of learning topology directly from data.

Our Position. In contrast to prior work, we propose a framework that jointly learns hypergraph structure and representation in an end-to-end differentiable manner. Beyond removing the reliance on static construction heuristics, our approach explicitly models hyperedge formation as a compatibility function rather than a similarity-based rule, improving the ability to recover heterogeneous cellular niches compared to similarity-based graph and hypergraph approaches.

3. HyperNiche Method

3.1. *Problem formulation*

Consider a spatial transcriptomics tissue sample containing N cells, represented as

$$\mathcal{D} = \{(\mathbf{x}_i, \mathbf{p}_i, y_i)\}_{i=1}^{N},$$

where $\mathbf{x}_i \in \mathbb{R}^d$ is the normalized gene-expression profile of cell i, $\mathbf{p}_i \in \mathbb{R}^2$ denotes its cell-centroid coordinates, and $y_i \in \{1, \ldots, C\}$ is its expert-curated cell-type label.

Our objective is to learn a function that predicts the cell-type label of each cell while simultaneously recovering localized, higher-order cellular organization. In contrast to a cell type, which describes the identity of an individual cell, we define a *cellular niche* as a spatially localized group of potentially heterogeneous cells whose joint organization may characterize a tissue microenvironment. Accordingly, cells with different expert-curated cell-type labels may belong to the same niche.

For each tissue sample, we construct a spatial candidate graph

$$\mathcal{G}_s = (\mathcal{V}, \mathcal{A}),$$

where $\mathcal{V} = \{1, \ldots, N\}$ is the set of cells and $(i, j) \in \mathcal{A}$ when cell i is included in the spatial k-nearest-neighbor candidate set of anchor cell j.

HyperNiche transforms the spatial candidate graph into a learned soft hypergraph

$$\mathcal{G}_h = (\mathcal{V}, \mathcal{E}_h, \mathbf{H}),$$

where $\mathcal{E}_h = \{e_1, \ldots, e_N\}$ is the set of candidate hyperedges. Each cell j serves as the anchor of one candidate hyperedge e_j; therefore, $|\mathcal{E}_h| = N$.

The incidence matrix $\mathbf{H} \in [0, 1]^{N \times N}$ represents the learned memberships, where H_{ij} denotes the membership strength of cell i in the hyperedge e_j anchored at cell j. Membership is permitted only when $(i, j) \in \mathcal{A}$; otherwise, $H_{ij} = 0$.

Let

$$\widehat{\mathbf{Y}} = F_\Theta(\mathbf{X}, \mathbf{P}, \mathcal{E}_s)$$

denote the cell-type predictions produced by HyperNiche, where $\mathbf{X} \in \mathbb{R}^{N\times d}$ is the expression matrix, $\mathbf{P} \in \mathbb{R}^{N\times 2}$ contains the spatial coordinates, and Θ contains the parameters of the encoder, hyperedge constructor, hypergraph message-passing layers, and classifier. The model is trained using expert-curated cell-type labels while learning the incidence weights jointly with the predictive representations.

Across a cohort containing multiple patients and tissue-microarray cores, the learning problem is evaluated using leave-one-patient-out cross-validation. In each outer fold, all cores belonging to one patient are held out together, and model parameters are learned using only the remaining patients. Thus, for a cohort with P unique patients, the evaluation contains P outer folds. Within each fold, one patient is reserved for testing, one of the remaining patients is reserved for validation, and the other $P - 2$ patients are used for model fitting.

Although cell-type supervision provides the training signal, HyperNiche is not intended solely as a cell-type classifier. The supervised objective guides the joint learning of cell representations and spatially localized hyperedge memberships. The classifier output is used to evaluate predictive performance, whereas the learned embeddings and hyperedges are used for representation-level comparison and structural characterization of the inferred niches, respectively.

3.2. *HyperNiche architecture*

Let $\mathbf{X} \in \mathbb{R}^{N\times d}$ denote the gene-expression matrix for a tissue sample containing N cells. HyperNiche first maps each cell to a latent representation using a multilayer perceptron:

$$\mathbf{H}^{(0)} = f_{\text{enc}}(\mathbf{X}), \tag{1}$$

where $f_{\text{enc}} : \mathbb{R}^d \to \mathbb{R}^{d_h}$ consists of two fully connected layers with ReLU activations and dropout. The resulting matrix $\mathbf{H}^{(0)} \in \mathbb{R}^{N\times d_h}$ contains the initial cell embeddings.

HyperNiche then learns a weighted incidence matrix $\mathbf{M} \in [0, 1]^{N\times N}$ over spatially permitted cell–anchor pairs. Each cell serves as the anchor of one candidate hyperedge, resulting in N candidate hyperedges. The entry M_{ij} represents the learned membership strength of cell i in the hyperedge anchored at cell j. A separate network assigns each anchored hyperedge a weight $w_j \in (0, 1)$, yielding the diagonal hyperedge-weight matrix

$$\mathbf{W} = \text{diag}(w_1, \dots, w_N). \tag{2}$$

Given the learned soft incidence matrix and hyperedge weights, HyperNiche propagates information from cells to hyperedges and back to cells using the normalized hypergraph message-passing operator introduced in spectral hypergraph learning[19] and subsequently adopted for hypergraph neural networks.[14] Specifically, a hypergraph-convolution layer is given by

$$\widetilde{\mathbf{H}}^{(\ell+1)} = \mathbf{D}_v^{-1/2}\mathbf{M}\mathbf{W}\mathbf{D}_e^{-1}\mathbf{M}^\top\mathbf{D}_v^{-1/2}\mathbf{H}^{(\ell)}\mathbf{\Theta}^{(\ell)}, \tag{3}$$

where $\mathbf{M}$ is the learned soft incidence matrix, $\mathbf{W}$ is the diagonal matrix of learned hyperedge weights, $\mathbf{D}_e$ and $\mathbf{D}_v$ are the hyperedge- and vertex-degree matrices, respectively, and $\mathbf{\Theta}^{(\ell)}$ is a learned linear transformation. The propagated representations are passed through a ReLU activation followed by layer normalization, which normalizes each cell representation across its feature dimensions and improves training stability:

$$\mathbf{U}^{(\ell+1)} = \text{LayerNorm}\left(\text{ReLU}\left(\widetilde{\mathbf{H}}^{(\ell+1)}\right)\right). \tag{4}$$

When the input and output feature dimensions agree, a residual connection combines the input representation with the normalized message-passing output:

$$\mathbf{H}^{(\ell+1)} = \mathbf{H}^{(\ell)} + \mathbf{U}^{(\ell+1)}. \tag{5}$$

This preserves the preceding cell representation while incorporating information propagated through the learned hypergraph. When the dimensions differ, the layer uses $\mathbf{H}^{(\ell+1)} = \mathbf{U}^{(\ell+1)}$.

After L hypergraph-convolution layers, a linear classifier maps the final cell representation to class logits:

$$\mathbf{Z} = \mathbf{H}^{(L)}\boldsymbol{\Theta}_{\text{cls}} + \mathbf{b}_{\text{cls}}, \tag{6}$$

where $\mathbf{Z} \in \mathbb{R}^{N\times C}$ and C is the number of reference cell types. The default implementation uses two hypergraph layers, a latent encoder dimension of 128, a hypergraph output dimension of 64, and dropout with probability 0.2.

3.3. *Hyperedge construction*

For each tissue sample, HyperNiche first constructs a sparse spatial candidate set

$$\mathcal{A} = \{(i,j) : i \in \mathcal{N}_k(j) \cup \{j\}\}, \tag{7}$$

where $\mathcal{N}_k(j)$ contains the k nearest spatial neighbors of anchor cell j. Nearest neighbors are determined from the raw physical cell-centroid coordinates using a KD-tree. Thus, each anchor has at most $k+1$ candidate members, including itself. Candidate memberships outside $\mathcal{A}$ are fixed to zero and are neither stored nor evaluated.

Raw cell-centroid coordinates are used to determine the spatial k-nearest-neighbor candidate pairs. Before being supplied to the learned compatibility function, the coordinates are normalized separately within each tissue core:

$$\widetilde{\mathbf{p}}_i = \frac{\mathbf{p}_i - \boldsymbol{\mu}_p}{a}, \tag{8}$$

where $\boldsymbol{\mu}_p$ is the mean coordinate vector of the core and a is the pooled standard deviation of the centered coordinate values across all cells and both spatial dimensions. For numerical stability, a is lower bounded by 10^{-6}. The same scalar is applied to both coordinate dimensions, thereby preserving their relative spatial scale.

For each permitted member–anchor pair $(i,j) \in \mathcal{A}$, HyperNiche constructs a relative spatial-feature vector containing the displacement of candidate member i from anchor j and their squared Euclidean distance:

$$\mathbf{s}_{ij} = \left[\widetilde{\mathbf{p}}_i - \widetilde{\mathbf{p}}_j \,\Big\|\, \|\widetilde{\mathbf{p}}_i - \widetilde{\mathbf{p}}_j\|_2^2\right], \tag{9}$$

where $\|$ inside the brackets denotes concatenation.

To distinguish the roles of a cell as a candidate member and as a hyperedge anchor, HyperNiche applies separate learned projections to the initial cell embeddings:

$$\mathbf{r}_i^{(m)} = \mathbf{W}_m \mathbf{h}_i^{(0)} + \mathbf{b}_m, \tag{10}$$

$$\mathbf{r}_j^{(a)} = \mathbf{W}_a \mathbf{h}_j^{(0)} + \mathbf{b}_a, \tag{11}$$

where $\mathbf{r}_i^{(m)}$ represents cell i in its candidate-member role and $\mathbf{r}_j^{(a)}$ represents cell j in its anchor role. Because the two roles use different learned projections, membership relationships are not required to be symmetric; in general, the compatibility of cell i with the hyperedge anchored at cell j may differ from that of cell j with the hyperedge anchored at cell i.

For each spatially permitted member–anchor pair $(i, j) \in \mathcal{A}$, HyperNiche computes an unnormalized compatibility score:

$$q_{ij} = \underbrace{\mathrm{Bilin}\left(\mathbf{r}_i^{(m)}, \mathbf{r}_j^{(a)}\right)}_{\text{member–anchor interaction}} + \underbrace{g_m\left(\mathbf{r}_i^{(m)}\right)}_{\text{member tendency}} + \underbrace{g_a\left(\mathbf{r}_j^{(a)}\right)}_{\text{anchor tendency}} + \underbrace{f_{\mathrm{sp}}\left(\mathbf{s}_{ij}\right)}_{\text{spatial compatibility}} . \tag{12}$$

Here, $\mathrm{Bilin}(\cdot, \cdot)$ is a learned bilinear compatibility function. The learned scalar-valued functions g_m and g_a capture, respectively, the general tendency of cell i to participate as a member and of cell j to admit neighboring cells into its anchored hyperedge. The spatial network $f_{\mathrm{sp}} : \mathbb{R}^{d_s} \to \mathbb{R}$, implemented as a two-layer multilayer perceptron, maps the relative spatial-feature vector $\mathbf{s}_{ij}$ to a scalar compatibility score. Together, these four terms allow hyperedge membership to depend on the learned representations and distinct roles of both cells, as well as their relative spatial configuration.

The soft incidence value is obtained independently for each permitted candidate pair:

$$M_{ij} = \begin{cases} 1, & i = j, \\ \sigma(q_{ij}), & (i, j) \in \mathcal{A},\ i \neq j, \\ 0, & (i, j) \notin \mathcal{A}, \end{cases} \tag{13}$$

where $\sigma(\cdot)$ denotes the sigmoid function. Fixing $M_{jj} = 1$ ensures that every anchored hyperedge contains its anchor.

HyperNiche also learns an importance weight for each hyperedge from its anchor representation:

$$w_j = \sigma\left(f_w\left(\mathbf{r}_j^{(a)}\right)\right), \tag{14}$$

where $f_w : \mathbb{R}^{d_r} \to \mathbb{R}$ is a two-layer multilayer perceptron that maps the anchor-role representation to a scalar logit, and σ is the sigmoid function. Thus, $w_j \in (0, 1)$ represents the learned importance of the hyperedge anchored at cell j. The learned hypergraph is characterized by the sparse soft incidence values $\{M_{ij} : (i, j) \in \mathcal{A}\}$ and the anchor-specific hyperedge weights $\{w_j\}_{j=1}^{N}$. No direct supervision is provided for individual hyperedge weights; instead, their learning signal arises from their influence on normalized hypergraph message passing and, consequently, downstream cell-type prediction.

3.4. *Learning objective and regularization*

HyperNiche is trained jointly for supervised cell-type classification and soft-hypergraph construction. Let $\mathcal{V}_{\mathrm{tr}}$ denote the labeled cells in the current training tissue core, and let α_c be the weight assigned to class c. The supervised component is the class-weighted cross-entropy loss

$$\mathcal{L}_{\mathrm{sup}} = -\frac{\displaystyle\sum_{i \in \mathcal{V}_{\mathrm{tr}}} \alpha_{y_i} \log\left(\frac{\exp(Z_{i,y_i})}{\sum_{c=1}^{C} \exp(Z_{i,c})}\right)}{\displaystyle\sum_{i \in \mathcal{V}_{\mathrm{tr}}} \alpha_{y_i}}, \tag{15}$$

where $Z_{i,c}$ is the logit predicted for cell i and class c, and y_i is the ground-truth class label of cell i. The denominator corresponds to the weighted-mean reduction used in the implementation.

The class weights are computed once for each cross-validation fold using all cells belonging to the training patients:

$$\alpha_c = \frac{n_{\mathrm{tr}}}{C n_c}, \tag{16}$$

where n_c is the number of cells assigned to class c among the training patients and $n_{\mathrm{tr}} = \sum_{c=1}^{C} n_c$. Thus, cells from less frequent classes receive greater weight in the supervised objective. For numerical stability, any zero class count is replaced by one when computing the weights.

Four regularizers constrain the learned incidence structure. First, an incidence-sparsity penalty discourages unnecessary non-anchor memberships:

$$\mathcal{L}_{\mathrm{sparse}} = \frac{1}{|\mathcal{A}_{\mathrm{off}}|} \sum_{(i,j) \in \mathcal{A}_{\mathrm{off}}} M_{ij}, \tag{17}$$

where $\mathcal{A}_{\mathrm{off}} = \{(i,j) \in \mathcal{A} : i \neq j\}$.

Second, a binary-entropy penalty encourages memberships to approach interpretable values near zero or one. For numerical stability, the off-diagonal memberships are clipped to $[10^{-4}, 1-10^{-4}]$ before the logarithms are evaluated. Let

$$\widehat{M}_{ij} = \mathrm{clip}(M_{ij}, 10^{-4}, 1 - 10^{-4}). \tag{18}$$

The binary-entropy penalty is

$$\mathcal{L}_{\mathrm{entropy}} = -\frac{1}{|\mathcal{A}_{\mathrm{off}}|} \sum_{(i,j) \in \mathcal{A}_{\mathrm{off}}} \left[\widehat{M}_{ij} \log \widehat{M}_{ij} + (1 - \widehat{M}_{ij}) \log(1 - \widehat{M}_{ij})\right]. \tag{19}$$

Third, let N denote the number of cells, and hence the number of anchored hyperedges, in the current tissue core. Define the soft size of the hyperedge anchored at cell j as

$$s_j = \sum_{i:(i,j) \in \mathcal{A}} M_{ij}. \tag{20}$$

A squared hinge penalty discourages degenerate hyperedges that are smaller than a prescribed minimum m_{min}:

$$\mathcal{L}_{\mathrm{deg}} = \frac{1}{N} \sum_{j=1}^{N} \max(0, m_{\mathrm{min}} - s_j)^2. \tag{21}$$

Finally, an overlap penalty discourages different anchored hyperedges from assigning membership mass to the same cells. Let

$$\overline{\mathbf{m}}_j = \frac{\mathbf{M}_{:j}}{\sum_i M_{ij} + \epsilon} \tag{22}$$

denote the normalized membership profile of hyperedge j, where $\epsilon > 0$ is a small constant used for numerical stability. For a set $\mathcal{S}$ of sampled anchors, the overlap penalty is

$$\mathcal{L}_{\text{overlap}} = \frac{1}{|\mathcal{S}|(|\mathcal{S}| - 1)} \sum_{\substack{j,r \in \mathcal{S} \\ j \neq r}} \overline{\mathbf{m}}_j^\top \overline{\mathbf{m}}_r. \tag{23}$$

For computational efficiency, the implementation randomly samples at most 256 anchors when evaluating this term.

The complete learning objective is

$$\mathcal{L} = \mathcal{L}_{\text{sup}} + \lambda_{\text{sparse}}\mathcal{L}_{\text{sparse}} + \lambda_{\text{entropy}}\mathcal{L}_{\text{entropy}} + \lambda_{\text{deg}}\mathcal{L}_{\text{deg}} + \lambda_{\text{overlap}}\mathcal{L}_{\text{overlap}}. \tag{24}$$

For the primary experiments, we set $\lambda_{\text{sparse}} = 10^{-3}$, $\lambda_{\text{entropy}} = 10^{-3}$, $\lambda_{\text{deg}} = 0.025$, and $\lambda_{\text{overlap}} = 10^{-3}$, with $m_{\min} = 4$. Model parameters are optimized using Adam with a learning rate of 10^{-3} and weight decay of 10^{-4}.

Because tissue cores define independent spatial graphs, the supervised and structural loss terms are evaluated separately for each training core. During each epoch, the training cores are processed in shuffled order, and one optimizer update is performed per core. Thus, no candidate memberships or hyperedges are constructed across tissue-core boundaries.

Unless otherwise stated, HyperNiche used $m_{\min} = 4$ and $\lambda_{\text{deg}} = 0.025$, where $m_{\min}$ denotes the minimum target soft-hyperedge size and λ_{deg} controls the degree-regularization term. This configuration served as the default for the primary comparison, ablation study, and structural characterization.

3.5. *Inference and niche-evaluation procedure*

For a held-out tissue sample, expression features are standardized using the feature-wise means and standard deviations estimated exclusively from the corresponding training patients. Spatial candidate pairs are constructed independently within each held-out tissue core from its raw cell-centroid coordinates. HyperNiche then performs a single forward pass to obtain the cell embeddings, class logits, soft incidence values, and hyperedge weights:

$$\left(\mathbf{Z}, \mathbf{H}^{(L)}, \mathbf{M}, \mathbf{w}\right) = F_\Theta\left(\mathbf{X}, \mathbf{P}, \mathcal{A}\right). \tag{25}$$

The checkpoint attaining the highest macro-averaged F1 score on the validation patient is used for evaluation of the held-out test patient. The predicted cell-type label for cell i is

$$\widehat{y}_i = \arg\max_{c \in \{1,\ldots,C\}} Z_{ic}. \tag{26}$$

The learned incidence values and hyperedge weights are retained as continuous outputs during inference. For downstream niche analysis, the hyperedge anchored at cell j is considered eligible when

$$w_j > \tau_w, \tag{27}$$

where τ_w denotes the hyperedge-weight threshold and is set to 0.05. At membership threshold τ_m, its discrete member set is

$$\widehat{e}_j^{(s)}(\tau_m) = \left\{ i : (i,j) \in \mathcal{A},\ M_{ij}^{(s)} > \tau_m \right\}, \tag{28}$$

where s denotes the model-initialization seed. An eligible hyperedge forms a valid seed-level niche only when

$$\left| \widehat{e}_j^{(s)}(\tau_m) \right| \geq 3. \tag{29}$$

Because the anchor membership is fixed at $M_{jj} = 1$, the anchor is included in this set and counts toward the minimum niche size. Seed-level descriptive analyses evaluate $\tau_m \in \{0.01, 0.02, 0.05\}$ to assess sensitivity to membership thresholding. The primary consensus and downstream biological analyses use $\tau_m = 0.10$.

Consensus niches are constructed separately within each held-out tissue core across five initialization seeds. For a given anchor j, both hyperedge eligibility and valid seed-level niche formation must occur in at least three of the five seeds. Membership votes are accumulated only from seeds in which the anchor produces a valid niche. A cell is included in the consensus niche when it belongs to the anchor's valid niche in at least three seeds:

$$\widehat{e}_j^{\text{cons}} = \left\{ i : \sum_{s=1}^{5} \mathbb{I}\left[i \in \widehat{e}_j^{(s)}(0.10) \ \wedge \ \left| \widehat{e}_j^{(s)}(0.10) \right| \geq 3 \right] \geq 3 \right\}. \tag{30}$$

The resulting consensus niche is retained only if $\left| \widehat{e}_j^{\text{cons}} \right| \geq 3$.

As a complementary stability analysis, seed-specific niches associated with the same anchor are compared pairwise using membership-set Jaccard similarity, continuous-membership cosine similarity, and cell-type-composition cosine similarity. Variability in niche size and spatial radius is also summarized across seeds. Consensus construction and stability analysis are applied only during downstream niche evaluation and do not alter model training or cell-type prediction.

3.6. *Structural niche measures*

We characterized each learned anchor-centered niche using its size, spatial extent, and cell-type composition. Let $M_{ij} \in [0,1]$ denote the learned membership of cell i in the hyperedge anchored at cell j, and let w_j denote the learned weight of that hyperedge. For the primary structural analysis, an anchor was eligible when $w_j > 0.05$, and its hard niche membership was defined as

$$\mathcal{N}_j = \{ i : M_{ij} > 0.10 \}.$$

Only niches containing at least three hard members were retained. Membership thresholds of 0.01, 0.02, and 0.05 were additionally considered in the threshold-sensitivity analysis.

Niche size was quantified in two ways. The soft niche mass was

$$S_j^{\text{soft}} = \sum_i M_{ij},$$

where the sum was taken over the candidate members of anchor j. The hard niche size was

$$S_j^{\mathrm{hard}} = |\mathcal{N}_j|.$$

The soft mass measures the total learned membership strength, whereas the hard size is the number of cells retained after thresholding.

Spatial extent was primarily summarized by the membership-weighted mean distance from the anchor:

$$R_j^{\mathrm{weighted}} = \frac{\sum_{i \in \mathcal{N}_j} M_{ij} \, \|\mathbf{s}_i - \mathbf{s}_j\|_2}{\sum_{i \in \mathcal{N}_j} M_{ij}},$$

where $\mathbf{s}_i$ and $\mathbf{s}_j$ are the raw centroid coordinates of the member and anchor cells, respectively. Since these raw coordinates are expressed in microns, all resulting distance metrics are correspondingly reported in microns. Lower values indicate greater spatial compactness. We additionally calculated the maximum anchor–member distance, the 90th percentile of the anchor–member distances, the membership-weighted radius of gyration around the niche center of mass, and the convex-hull area. For comparison, maximum and 90th-percentile radii were also calculated for a spatial k-nearest-neighbor neighborhood containing the same number of cells as the learned niche.

Cell-type composition was summarized using normalized Shannon entropy. For a niche containing proportions p_{jc} of reference cell type c, the normalized composition entropy was

$$E_j = -\frac{\sum_{c=1}^{C} p_{jc} \log p_{jc}}{\log C},$$

where C was the number of reference cell types present in the corresponding tissue core. Terms with $p_{jc} = 0$ contributed zero. Entropy approached zero when a niche was dominated by one cell type and approached one when its members were distributed uniformly across all reference cell types represented in the core. This measure describes compositional heterogeneity and was not interpreted as direct evidence of biological validity.

Structural measures were first averaged across eligible niches within each tissue core and initialization seed. Seed-level summaries were then averaged within each core, followed by aggregation across cores belonging to the same patient. This hierarchical aggregation prevented tissue cores containing more cells or more inferred niches from contributing disproportionately to the patient-level summary. Optimization variability across seeds and variability across cores were retained as separate descriptive quantities. Niches failing the hyperedge-weight, membership, or minimum-size criteria were excluded rather than assigned zero-valued structural measurements.

4. Experimental Methods

4.1. *Study objectives and evaluation design*

We designed the experiments to answer four questions: (1) whether HyperNiche recovers tissue organization more accurately than established spatial-domain and hypergraph methods; (2) what spatial and compositional properties characterize the learned neighborhoods, including their compactness, cell-type selectivity, and capacity to include multiple cell types; (3) which model components contribute to these behaviors; and (4) whether the conclusions remain stable across patients, tissue cores, random seeds, and reasonable hyperparameter choices.

All preprocessing choices, model-selection rules, and evaluation metrics were fixed before evaluation on held-out tissue units. We used leave-one-patient-out (LOPO) evaluation, with one outer fold for each unique patient in a dataset. Within each outer fold, one patient was held out for testing, one of the remaining patients was designated for validation, and all other patients were used for model fitting. Patients were ordered by identifier, and the patient following the test patient in this fixed ordering, with cyclic wraparound, served as the validation patient. Consequently, each patient served once as the test patient and once as the validation patient across the outer folds. All tissue cores and samples from a patient remained in the same partition.

The complete fold assignments were generated once and held fixed across the five random seeds (0, 7, 42, 123, and 999). The validation patient was excluded from model fitting and was used only for checkpoint and prespecified hyperparameter selection; the outer test patient was used only for final evaluation. Thus, the procedure comprised outer LOPO evaluation with a deterministic patient-level validation split, rather than exhaustive nested cross-validation.

HyperNiche was trained using reference cell-type labels, but its classifier, cell embeddings, and learned hyperedges were evaluated for distinct purposes. Direct classifier performance was assessed using macro-F1 and balanced accuracy. To compare representation quality across methods with different native outputs, we applied the same K-means postprocessing protocol to each method's embeddings and measured agreement with the reference annotations using ARI and NMI. The learned hyperedges were evaluated separately through structural and compositional niche characterization. Thus, ARI and NMI measure label alignment within the embedding space and are not substitutes for direct classification performance or independent biological validation.

4.2. *Spatial transcriptomics cohorts*

We evaluated HyperNiche on two 10x Genomics Xenium cohorts from GSE308148:[20] non-small cell lung cancer (NSCLC) and breast cancer (BrC). Each cohort comprises multiple tissue-microarray (TMA) cores from multiple patients, permitting patient-level evaluation across spatially distinct tissue units rather than random cell-level splitting. Table 1 summarizes the cohort characteristics after quality control.

Cells with fewer than 10 detected transcripts were removed, as were genes detected in fewer than five cells. Expression counts were normalized independently for each cell to a total of 10^4 counts and subsequently transformed using $\log(1 + x)$. The resulting gene-expression matrix was used as the node-feature matrix for HyperNiche. Within each cross-validation fold, feature-wise means and standard deviations were estimated using the training patients and applied to the training, validation, and held-out test samples.

Cell-centroid coordinates were retained in their native spatial coordinate system and used to construct spatial neighborhoods through a k-nearest-neighbor graph. Expression-derived principal components were not used to define spatial relationships.

Table 1. Characteristics of the spatial transcriptomics cohorts after quality control.

Cohort	Patients	Cores	Pre-QC cells	Post-QC cells	Genes
NSCLC (lung cancer)	5	24	56,107	53,473	536
Breast cancer	4	18	30,917	30,704	536

Reference annotations. Reference cell-type annotations were provided by an external group and were established independently of HyperNiche before model fitting. The annotations were assigned manually to expression-derived cell clusters using established marker genes available in the targeted Xenium panels, and each cell inherited the label of its cluster. These labels were used as supervised targets and as the reference partition for evaluation.

4.3. *Preprocessing and model fitting*

The PCA and Harmony embeddings used for expression-based clustering and manual cell annotation were not provided as inputs to HyperNiche. Instead, HyperNiche used the preprocessed gene-expression feature matrix stored in the primary `X` field of the AnnData object. Within each patient-level cross-validation fold, feature-wise means and standard deviations were estimated exclusively from the training tissue cores and used to standardize the training, validation, and test features.

For each anchor cell, HyperNiche restricted candidate hyperedge membership to the anchor and its 20 nearest spatial neighbors within the same tissue core. Candidate neighborhoods were constructed from the raw cell-centroid coordinates using a k-d tree. Because every cell served as an anchor, a tissue core containing N cells produced N candidate hyperedges, each containing at most 21 candidate members, including the anchor. Cell-centroid coordinates, expressed in microns (μm), were centered within each tissue core and divided by a single coordinate-scale factor before being supplied to the learned spatial module.

The node encoder was a two-layer multilayer perceptron with a hidden dimension of 128, ReLU activations, and dropout of 0.2 after the first layer. Separate linear projections mapped the encoded cell features to 64-dimensional member and anchor representations. Membership logits combined a bilinear member–anchor compatibility term, role-specific member and anchor biases, and a spatial term computed by a two-layer multilayer perceptron from the relative coordinate features $(\Delta x, \Delta y, d^2)$. Hyperedge weights were predicted from the anchor representation using a two-layer multilayer perceptron with a sigmoid output. The learned memberships and hyperedge weights were subsequently used by two hypergraph-convolution layers with an output dimension of 64.

Models were trained on complete tissue cores, with one core per loader batch, using Adam with a learning rate of 10^{-3}, weight decay of 10^{-4}, and a maximum of 100 epochs. The supervised loss used class weights computed exclusively from the training cells. Unless otherwise varied in the sensitivity analysis, the minimum target soft hyperedge size was 4, and the sparsity, entropy, degeneracy, and overlap regularization weights were 10^{-3}, 10^{-3}, 0.025, and 10^{-3}, respectively. All 100 epochs were completed, and, separately for each random seed, the checkpoint attaining the highest macro-averaged F1 score on the fold's designated validation patient was selected. This checkpoint was then evaluated once on the outer held-out test patient. The five initialization seeds used the same patient-level training, validation, and test assignments.

The framework was implemented in Python 3.10 using PyTorch 2.6. AnnData, SciPy 1.15.3, and scikit-learn 1.2.2 were used for data preprocessing, spatial-neighborhood construction, clustering, and downstream evaluation. All experiments were conducted on the University of Houston Carya HPC cluster (HPE-DSI), using nodes equipped with dual Intel Xeon Gold 6252 processors and NVIDIA V100 Tensor Core GPUs.

4.4. *Comparison methods*

We compared HyperNiche with two spatial-transcriptomics methods, SpaGCN and GraphST, and two general-purpose hypergraph neural networks, HyperGCN and HyperSTAR. A static k-nearest-neighbor construction with fixed neighborhoods was evaluated separately as an ablation control.

All methods received the same quality-controlled cells and used identical patient-level training, validation, and test assignments, including the same designated validation patient in each outer fold. Each method was fitted using only the training patients. When hyperparameter or checkpoint selection was required, selection used only the designated validation patient and the prespecified method-specific search budget. The outer test patient was not used for hyperparameter tuning, checkpoint selection, or any other model-selection decision.

HyperNiche used reference cell-type labels in its supervised objective, whereas the comparison methods were trained using their native unsupervised objectives without access to these labels. Reference annotations were used only for their post-training evaluation. Thus, the common K-means protocol standardizes downstream representation-level evaluation but does not provide a supervision-matched comparison.

4.5. *Outcome measures and statistical analysis*

Macro-F1 and balanced accuracy were the primary measures of direct classifier performance. ARI and NMI were complementary representation-level measures obtained by applying a common K-means protocol to the learned embeddings. Checkpoint selection used validation macro-F1 exclusively and did not use test-set ARI, NMI, or structural outcomes.

Where patient- and seed-indexed outputs were available, metrics were computed for each held-out tissue core and seed, averaged first across cores within each patient and then across seeds, and summarized across patients using the mean and sample standard deviation. Seeds, tissue cores, and cells were not treated as independent biological replicates. Table 2 reports aggregate point estimates from the available benchmark outputs.

Learned hyperedges were characterized separately using niche size, physical radius, cell-type-composition entropy, cross-seed membership and compositional stability, local and global cell-type enrichment, spatial compactness by radius and area, and nearby-niche membership overlap. Their definitions, aggregation procedures, directionality, and treatment of undefined values are provided in Section 3.6; consensus-niche construction and permutation procedures are described in Section 5.5. These outcomes were summarized descriptively and provide evidence of structural consistency and biological plausibility rather than independent experimental validation. They were not treated as independent biological replicates or used to redefine the primary predictive conclusions.

Silhouette and Davies–Bouldin scores were computed within each applicable method's embedding space and interpreted only as descriptive measures of within-method latent-space separation.

4.6. *Ablation and sensitivity analyses*

We evaluated six component ablations while preserving the input data, patient-level partitions, preprocessing, training budget, checkpoint-selection rule, and evaluation procedure used for the full HyperNiche model. The symmetric-similarity ablation replaced the asymmetric anchor–member compatibility mechanism with a symmetric inner-product similarity function. The static spatial k-nearest-neighbor (kNN) ablation replaced learned hyperedge membership with fixed, distance-defined neighborhoods. The remaining ablations: (i) assigned unit weight to every hyperedge instead of learning hyperedge importance; (ii) removed the learned spatial-logit term while retaining the spatial kNN candidate graph; (iii) removed the degeneracy penalty by setting $\lambda_{\text{deg}} = 0$; and (iv) removed hypergraph message passing so that cell-type predictions were generated directly from the encoded cell features. Each ablation was designed to isolate one model component while leaving the remaining architecture and training protocol unchanged.

Sensitivity analyses varied prespecified structural and optimization parameters while retaining the same patient-level partitions and evaluation procedure. The analyzed factors included the minimum target soft hyperedge size ($m_{min} \in \{2.0, 3.0, 4.0\}$), degeneracy-regularization weight ($\lambda_{deg} \in \{0.005, 0.01, 0.025, 0.05, 0.1\}$), spatial candidate-neighborhood size $k \in \{8, 12, 20\}$, membership threshold, and random initialization seed. Parameter sensitivity was evaluated separately from component ablation: sensitivity analyses examined the robustness of the full model to reasonable parameter choices, whereas ablations tested whether particular learned components were necessary.

4.7. *Biological and structural characterization of learned niches*

We characterized consensus niches using cross-seed reproducibility, cell-type enrichment, and spatial compactness. All analyses used the primary hard-membership threshold of 0.10. Across five random seeds, an anchor was considered majority eligible if its learned hyperedge weight exceeded 0.05 in at least three seeds. A consensus niche was retained if the anchor formed a valid niche in at least three seeds and contained at least three assigned cells in at least three seeds. Cohort-level consensus recovery was calculated as the number of retained consensus niches divided by the number of majority-eligible anchors.

For each consensus niche, cell-type enrichment was evaluated using two size-matched permutation null models with 1,000 samples. The local null sampled the same number of cells from within 1.5 times the observed niche radius around its anchor. The global spatial null selected a random anchor from the same tissue core and used its k-nearest spatial neighbors, where k equaled the observed niche size. One-sided permutation tests assessed whether the observed count of each represented cell type exceeded its corresponding null distribution. A niche was classified as locally or globally enriched if at least one cell type remained significant after false-discovery-rate correction.

Spatial compactness was evaluated using the same size-matched local sampling region. Radius was defined as the maximum Euclidean distance from the anchor to an assigned member, whereas area was defined as the convex-hull area of the member-cell coordinates. A niche was classified as compact by radius or area if the corresponding observed statistic was significantly smaller than its permutation-null distribution after false-discovery-rate correction.

Cross-seed membership reproducibility was measured using the Jaccard similarity between the cell sets assigned to the same anchor in different seed runs. Compositional stability was measured using the cosine similarity between the corresponding cell-type proportion vectors. Both measures were summarized descriptively across valid consensus niches. These analyses provide descriptive evidence of structural consistency and biological plausibility rather than independent experimental validation of the learned niches.

4.8. *Reproducibility*

The released repository will include patient and tissue-core identifiers, mappings between cores and patients, outer-fold assignments, analysis-ready input specifications, configuration files and initialization seeds for every comparison method and ablation, software and dependency versions, and executable scripts for reproducing the reported model-training results, tables, and figures. The reference cell-type annotations used in this study were provided by an external collaborating group and were generated independently of HyperNiche. Their annotation workflow is outside the scope of the present software release.

The repository will document the source and version of each dataset, the expected input-data schema, and the commands required to run model training and evaluation. Data or annotations that cannot be redistributed will be accompanied, where applicable, by accession information or instructions for obtaining authorized access.

All methods used the patient-level fold assignments defined in Section 4.1. Within each fold, feature-standardization parameters were estimated exclusively from the training tissue cores and applied unchanged to the validation and test cores. Test labels and test-set performance were not used for early stopping, hyperparameter selection, checkpoint selection, or any other model-selection decision. The released fold assignment file identifies the training, validation, and test patients used in every outer fold.

5. Results

5.1. *Comparison with existing methods*

Table 2 reports a representation-level comparison using a common K-means evaluation protocol. HyperNiche achieved the highest ARI and NMI on both cohorts, indicating that its supervised embeddings were more strongly aligned with the reference cell-type annotations than those produced by the comparison methods. Because HyperNiche used these annotations during training whereas the unsupervised baselines did not, these results demonstrate task-aligned representation quality but should not be interpreted as a supervision-matched comparison of unsupervised clustering ability.

Table 2. **Representation-level comparison on the breast- and lung-cancer cohorts.** For each method, K-means was applied to its learned embeddings, and agreement with the reference cell-type annotations was measured using ARI and NMI. Methods were trained using their native objectives; HyperNiche used cell-type supervision, whereas the comparison methods did not use reference labels during representation learning. The shared clustering protocol standardizes post-training evaluation but does not constitute a supervision-matched comparison. Silhouette and Davies–Bouldin scores were computed within each method's own embedding space and are interpreted descriptively.

	Annotation Agreement		Latent Space Quality	
Method	**ARI ↑**	**NMI ↑**	**Silhouette ↑**	**Davies–Bouldin ↓**
Dataset 1: NSCLC Cohort				
SpaGCN	0.327	0.407	0.116	2.322
GraphST	0.160	0.220	0.147	2.144
HyperGCN	0.224	0.398	**0.346**	**0.879**
HyperSTAR	0.093	0.170	0.312	1.110
HyperNiche (Ours)	**0.428**	**0.474**	0.319	1.318
Dataset 2: BrC Cohort				
SpaGCN	0.251	0.335	0.138	2.208
GraphST	0.209	0.260	0.220	1.783
HyperGCN	0.175	0.244	0.369	**0.880**
HyperSTAR	0.121	0.174	**0.398**	1.033
HyperNiche (Ours)	**0.382**	**0.449**	0.395	1.075

HyperNiche achieved the highest annotation agreement in both cohorts. In the lung cancer cohort, it obtained an ARI of 0.428 and an NMI of 0.474. Relative to the strongest competing result for each metric, these correspond to absolute improvements of 0.101 in ARI over SpaGCN and 0.067 in NMI over SpaGCN. In the breast cancer cohort, HyperNiche obtained an ARI of 0.382 and an NMI of 0.449, improving over SpaGCN by 0.131 and 0.114, respectively. These results indicate that the cell representations learned by HyperNiche align more closely with the manual cell-type annotations under the common clustering protocol.

The intrinsic clustering indices present a more nuanced pattern. In lung cancer, HyperGCN achieved the highest Silhouette score (0.346) and the lowest Davies–Bouldin index (0.879), while HyperNiche obtained 0.319 and 1.318, respectively. In breast cancer, HyperNiche's Silhouette score of 0.395 was close to the highest value of 0.398 achieved by HyperSTAR, whereas HyperGCN obtained the lowest Davies–Bouldin index (0.880). Thus, greater geometric compactness within a learned latent space did not necessarily translate into greater agreement with the expert annotations. HyperNiche's principal advantage in this comparison is its consistently stronger recovery of the annotated cell-type partition, rather than uniformly optimal performance on every intrinsic separation measure.

ARI and NMI quantify agreement with the reference cell-type partition but do not, by themselves, establish that a learned neighborhood constitutes a biologically meaningful cellular niche. Similarly, Silhouette and Davies–Bouldin values are calculated within method-specific embedding spaces and should therefore be interpreted as descriptive measures rather than as direct evidence of biological validity. We therefore additionally examined the reproducibility, compositional selectivity, local enrichment, and spatial extent of the niches learned by HyperNiche.

Table 3. **Structural and compositional properties of the spatial niches learned by HyperNiche.** For each patient, niche-level measurements were first averaged across consensus niches; values are reported as the mean±SD of these patient-level summaries across patients. Membership Jaccard measures cross-seed reproducibility, niche size is the number of hard-assigned cells, normalized entropy measures compositional heterogeneity, and spatial radius measures spatial extent. For locally enriched (%), the percentage of locally enriched consensus niches was calculated within each patient before summarization across patients. Lower normalized entropy indicates greater compositional selectivity.

Cohort	Membership Jaccard ↑	Niche size	Norm. entropy ↓	Locally enriched (%) ↑	Spatial radius
Lung cancer	0.859 ± 0.127	9.79 ± 2.17	0.087 ± 0.050	75.4 ± 21.0	18.50 ± 2.30
Breast cancer	0.792 ± 0.090	11.39 ± 1.65	0.119 ± 0.062	43.5 ± 15.8	20.88 ± 4.08

As shown in Table 3, the learned niches were reproducible across training seeds, with mean membership Jaccard values of 0.859 in lung cancer and 0.792 in breast cancer. Their low normalized entropy values (0.087 and 0.119, respectively) indicate compositionally selective rather than uniformly mixed memberships. The niches were also spatially localized, containing approximately 10–11 cells on average and having mean spatial radii of 18.50μm and 20.88μm.

Local enrichment was observed for a mean of 75.4% of consensus niches in lung cancer and 43.5% in breast cancer. Thus, enrichment was less prevalent and more variable in the breast cancer cohort, although a substantial fraction of its learned niches remained locally enriched. Collectively, these results provide complementary evidence that HyperNiche recovers reproducible, compositionally selective, and spatially localized cellular neighborhoods. They should nevertheless be interpreted as descriptive evidence of biological plausibility rather than proof that every inferred niche represents a distinct biological program. Reported standard deviations describe variation across patients; repeated training seeds were used to construct the consensus niches and were not treated as independent biological replicates.

5.2. *Ablation study*

We evaluated six ablations to determine how the principal components of HyperNiche contribute to predictive performance and learned niche structure. Each ablation modified only the indicated component while preserving the patient-level folds, five random seeds, training budget, checkpoint-selection procedure, and evaluation protocol. Predictive metrics were averaged across seeds within each held-out patient and then summarized across the four patients. Structural evaluation considered consensus-niche formation, cross-seed membership agreement, niche size, composition entropy, and local cell-type enrichment.

Table 4. **Ablation results on the breast cancer cohort.** Values are mean±SD across four held-out patients after averaging five seeds within each patient. Δ denotes the change relative to full HyperNiche.

Model	Macro-F1	ΔF1	Bal. Acc.	ΔBA
Full HyperNiche	0.664 ± 0.072	—	0.651 ± 0.076	—
Symmetric-similarity constructor	0.597 ± 0.086	-0.067	0.597 ± 0.078	-0.054
Static spatial kNN	0.249 ± 0.166	-0.414	0.316 ± 0.187	-0.335
Unit hyperedge weights	0.638 ± 0.076	-0.026	0.637 ± 0.077	-0.014
No learned spatial logit	0.664 ± 0.075	$+0.001$	0.653 ± 0.078	$+0.002$
No degeneracy penalty	0.672 ± 0.076	$+0.008$	0.658 ± 0.078	$+0.007$
No message passing	0.675 ± 0.067	$+0.012$	0.658 ± 0.072	$+0.008$

Table 5. **Effects of ablations on learned niche structure.** Enrichment is reported as the range of patient-level fractions of locally enriched consensus niches. NE indicates not estimable because no consensus niches were recovered.

Model	Consensus	Hard size	Jaccard	Entropy	Locally enriched
Full HyperNiche	11,160	11.39	0.792	0.119	24–63%
Symmetric similarity	24,819	12.35	0.722	0.277	27–46%
Static spatial kNN	30,704	21.00	1.000	0.491	0%
Unit hyperedge weights	16,736	9.59	0.645	0.155	13–46%
No learned spatial logit	11,684	11.19	$\sim$0.770	0.110	22–64%
No degeneracy penalty	0	NE	NE	NE	NE
No message passing	30,704	21.00	1.000	0.491	0%

Replacing the learned memberships and hyperedge weights with a static spatial kNN hypergraph produced the largest performance reduction, decreasing Macro-F1 from 0.664 ± 0.072 to 0.249 ± 0.166 and balanced accuracy from 0.651 ± 0.076 to 0.316 ± 0.187. The resulting deterministic 21-cell neighborhoods exhibited no significant local cell-type enrichment. Restricting the niche constructor to symmetric similarity also reduced performance consistently across all held-out patients (Macro-F1: 0.597 ± 0.086) and produced less stable, compositionally less focused niches. Replacing learned hyperedge importance with unit weights caused a smaller but consistent reduction (Macro-F1: 0.638 ± 0.076), accompanied by lower membership stability and enrichment.

Two ablations preserved predictive accuracy but substantially altered niche formation. Without the degeneracy penalty, Macro-F1 remained comparable to the full model (0.672 ± 0.076), but no consensus niches were recovered, indicating that, under the evaluated configuration, this regularizer is necessary for reproducible niche construction rather than classification alone. Removing hypergraph message passing similarly preserved prediction (Macro-F1: 0.675 ± 0.067), but every eligible anchor retained its complete 21-cell candidate neighborhood. These perfectly reproducible yet nonselective neighborhoods showed no significant local enrichment, suggesting that message passing helps couple membership learning to the supervised objective.

Finally, removing the learned spatial-logit term had negligible effects on prediction (Macro-

Table 6. **Sensitivity of HyperNiche to the minimum niche size $m_{\min}$ and degeneracy penalty $\lambda_{\deg}$.** The locally enriched column reports the range across held-out patients. NE indicates not evaluable because no valid cross-seed consensus niches were recovered. All configurations were evaluated on the same BrC cohort and candidate anchor population. Consensus niches are therefore reported as raw counts for descriptive comparison across parameter settings; the counts are not intended for comparison across cohorts.

$m_{\min}$	$\lambda_{\deg}$	Macro-F1	Bal. Acc.	Consensus (n)	Jaccard	Hard size	Norm. entropy	Radius	Enriched
2	0.005	0.674±0.075	0.659±0.077	0	NE	NE	NE	NE	NE‡
4	0.010	0.671±0.074	0.658±0.079	7,264	0.795±0.078†	10.40±2.28†	0.091±0.033†	18.72±3.56†	37–61%†
4	0.025	0.657±0.079	0.651±0.081	12,029	0.783±0.076	11.08±1.55	0.117±0.057	21.44±4.76	22–65%
3	0.050	0.665±0.076	0.654±0.077	11,437	0.766±0.100	11.03±1.52	0.112±0.058	21.97±5.96	19–65%
4	0.050	0.653±0.070	0.650±0.076	11,860	0.712±0.176	10.87±2.02	0.128±0.068	20.07±2.91	15–61%
4	0.100	0.650±0.071	0.646±0.077	9,262	0.704±0.147	10.95±2.37	0.145±0.065	21.16±3.14	26–57%

† Structural summaries use the three evaluable patients; no valid consensus niches were recovered for the fourth patient. ‡ No valid cross-seed consensus niches were recovered for any held-out patient.

F1: 0.664 ± 0.075) or the principal structural properties. Because spatial proximity continued to determine candidate neighborhoods, this result suggests that explicit coordinate-dependent scoring provided limited additional information beyond spatial candidate selection and expression-based compatibility in this cohort. Collectively, the ablations show that learned selective memberships provide the largest predictive benefit, while asymmetric compatibility, hyperedge weighting, message passing, and degeneracy regularization contribute complementary improvements in niche specificity, reproducibility, and biological enrichment.

5.3. *Sensitivity to niche-regularization parameters*

We evaluated the sensitivity of HyperNiche to the minimum niche size $m_{\min}$ and degeneracy-penalty weight $\lambda_{\deg}$. Experiments used the same patient-level cross-validation protocol as the primary evaluation, with five random seeds per held-out patient. Predictive metrics were first averaged across seeds within each patient and are reported as the mean ± sample standard deviation across the four held-out patients. Structural metrics were computed from cross-seed consensus niches and similarly summarized at the patient level.

Table 6 summarizes the sensitivity of consensus-niche recovery and structural characterization to the minimum niche size and degeneracy penalty in the BrC cohort. Because every configuration was evaluated on the same cohort and candidate anchor population, the raw consensus-niche counts provide a descriptive comparison across parameter settings. Predictive performance was comparatively insensitive to the tested niche-regularization parameters. Across all six configurations, Macro-F1 varied from 0.650 to 0.674, and balanced accuracy varied from 0.646 to 0.659. In contrast, the reproducibility and structural quality of the learned niches were substantially more sensitive to the degeneracy-penalty weight $\lambda_{\deg}$.

At $\lambda_{\deg} = 0.005$, HyperNiche maintained predictive performance but failed to recover any valid cross-seed consensus niches for all four held-out patients. Increasing $\lambda_{\deg}$ to 0.010 partially restored niche formation; however, one held-out patient still produced no valid consensus niches. Accordingly, the structural statistics at this setting were computed using only the three evaluable patients and should not be interpreted as evidence of uniformly superior niche

Table 7. **Sensitivity of HyperNiche to the spatial neighborhood size** k, with minimum niche size 4 and $\lambda_{deg} = 0.025$. Values are the mean $\pm$ standard deviation across five random seeds and describe optimization, not patient-level, variability.

k	Macro-F1	Balanced accuracy	Macro-precision
8	0.6539 ± 0.0084	0.6405 ± 0.0061	$\mathbf{0.7480 \pm 0.0204}$
12	0.6547 ± 0.0103	0.6477 ± 0.0089	0.7371 ± 0.0148
20	$\mathbf{0.6570 \pm 0.0105}$	$\mathbf{0.6526 \pm 0.0116}$	0.7353 ± 0.0171

quality.

Intermediate penalty values, $\lambda_{\text{deg}} \in \{0.025, 0.050\}$, produced valid consensus niches for all four held-out patients while preserving similar predictive performance. The configuration $m_{\min} = 4, \lambda_{\text{deg}} = 0.025$ recovered the largest number of consensus niches (12,029) and exhibited strong cross-seed membership stability (0.783 ± 0.076). At $\lambda_{\text{deg}} = 0.050$, increasing $m_{\min}$ from 3 to 4 had little effect on predictive performance or mean niche size, but decreased membership Jaccard from 0.766 ± 0.100 to 0.712 ± 0.176 and increased normalized composition entropy from 0.112 ± 0.058 to 0.128 ± 0.068.

Increasing λ_{deg} further to 0.100 retained valid consensus niches for all patients but reduced their total number to 9,262, decreased membership stability to 0.704 ± 0.147, and increased normalized entropy to 0.145 ± 0.065. Thus, an excessively weak penalty failed to support reproducible niche formation, whereas an excessively strong penalty reduced niche stability and compositional specificity. Overall, the results identify $\lambda_{\text{deg}} \in [0.025, 0.050]$ as the most reliable structural operating range, while demonstrating that HyperNiche's cell-type prediction performance remains robust across the tested settings. The default configuration, $m_{\min} = 4$ and $\lambda_{\text{deg}} = 0.025$, was used for the primary analyses. The remaining configurations were evaluated only to assess sensitivity to the niche-regularization parameters.

5.4. *Sensitivity to spatial neighborhood size*

We evaluated the sensitivity of HyperNiche to the spatial neighborhood size k, which determines the number of spatial neighbors considered during initial neighborhood construction. Holding the minimum niche size and regularization coefficient fixed at 4 and $\lambda_{\text{deg}} = 0.025$, respectively, we evaluated $k \in \{8, 12, 20\}$ using the same leave-one-patient-out folds and five matched random seeds. Table 7 reports the mean and standard deviation across the five seed-specific cohort-level performance estimates. This sensitivity analysis is descriptive: each seed produced one cohort-level performance estimate, and seeds were not treated as biological replicates or used for statistical inference.

Performance remained stable across the three neighborhood sizes. Relative to $k = 8$, increasing the neighborhood size to $k = 20$ changed Macro-F1 by only $+0.0032$, while balanced accuracy increased by 0.0121 and macro-precision decreased by 0.0126. The balanced-accuracy improvement at $k = 20$ was observed for all four held-out patients; however, the patient- and seed-level comparisons were mixed across the other metrics. Thus, the results do not indicate that a single value of k is uniformly optimal. Instead, they demonstrate that HyperNiche is robust to a substantial variation in the spatial neighborhood size, with only modest trade-offs between recall-oriented balanced accuracy and macro-precision.

5.5. ***Structural and biological characterization of learned niches***

We characterized the consensus niches using the reproducibility, compositional-stability, enrichment, spatial-compactness, and nearby-overlap procedures defined in Section 5.5. Results were summarized separately for the breast- and lung-cancer cohorts, with patients treated as the independent biological units.

Across the breast-cancer cohort, HyperNiche recovered 11,160 consensus niches from 18 tissue-microarray cores belonging to four patients. Of the anchors eligible in a majority of seeds, 97.3% yielded a consensus niche. Membership Jaccard similarity ranged from 0.703 to 0.914 across patients, and composition cosine similarity ranged from 0.971 to 0.998, indicating strong stability across random initializations. Moreover, 97.2% and 93.4% of consensus niches were significantly compact by maximum radius and convex-hull area, respectively. A total of 40.8% were significantly enriched for at least one cell type relative to their immediate local context. None remained significant against the more stringent global spatial-neighborhood null; thus, the breast-cancer results support local cell-type selectivity, but not stronger enrichment than conventional spatial neighborhoods throughout the tissue.

The lung-cancer cohort contained 15,198 consensus niches across 24 cores from five patients. Consensus reproducibility was 94.3%, with patient-level membership Jaccard similarity ranging from 0.653 to 0.967 and composition cosine similarity ranging from 0.900 to 1.000. Most niches were spatially compact: 93.6% were significant by maximum radius and 80.0% by convex-hull area. Lung niches showed stronger compositional selectivity than breast niches: 70.5% were significantly enriched relative to their immediate local context, and 10.0% remained significant relative to matched spatial k-nearest-neighbor neighborhoods. Nearby learned niches had low mean membership overlap in both cohorts, supporting limited local redundancy.

Figure 1 summarizes the complementary cohort-level patterns. Breast cancer provides particularly strong evidence of cross-seed reproducibility and spatial compactness, whereas lung cancer provides stronger evidence of cell-type selectivity. These results should not be interpreted as implying that every learned niche is heterotypic: many niches are dominated by one annotated cell type. Rather, they show that HyperNiche can recover stable and localized higher-order structures without requiring cell-type homogeneity, and that a substantial subset is selectively composed relative to its spatial background.

5.5.1. *Representative tissue cores*

For qualitative and core-level analyses, we selected one representative core from each cohort using a prespecified combination of niche count, reproducibility, cell-type coverage, local enrichment, compactness, and low redundancy, rather than choosing the core with the single most favorable metric. Breast patient 2, core 96 contained 1,384 consensus niches, with 97.1% reproducibility, 67.7% local enrichment, and 97.8% radius-based compactness. At seed 42, 72.7% of its niches contained at least two annotated cell types and all eight breast-cancer cell types were represented. Lung patient 1, core 119 contained 489 consensus niches, with

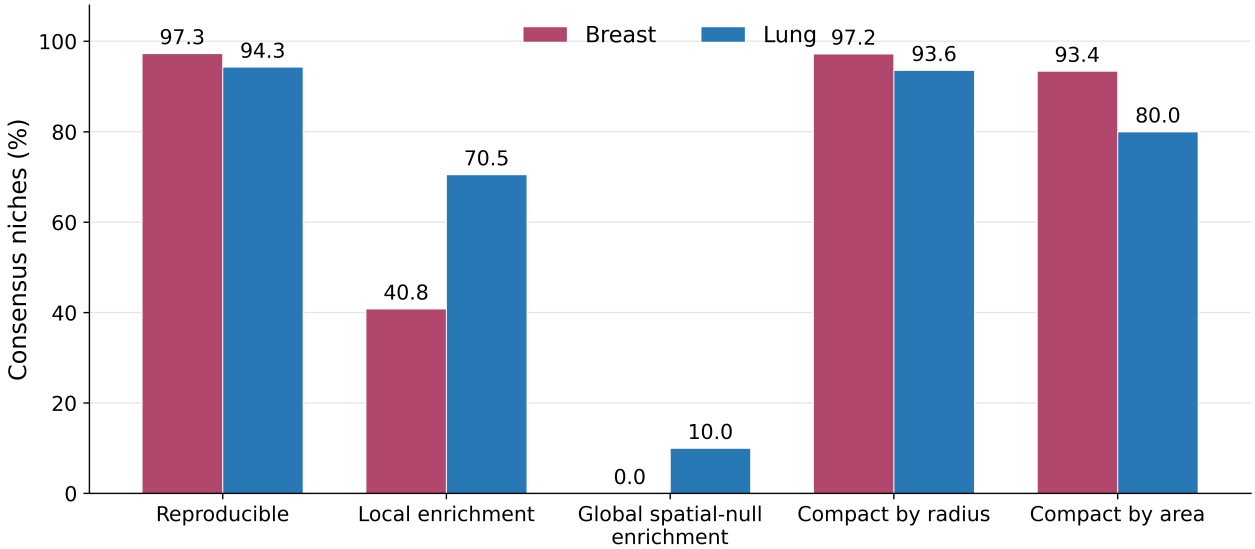

Fig. 1. Biological characterization of consensus niches in the breast- and lung-cancer cohorts. Reproducibility denotes the fraction of majority-eligible anchors yielding a consensus niche. Enrichment was tested relative to either the immediate local context or matched ordinary spatial neighborhoods. Compactness was evaluated by maximum radius and convex-hull area. Percentages summarize niches, whereas patients—not individual niches—constitute the independent biological units.

93.9% reproducibility, 72.8% local enrichment, and 94.7% radius-based compactness. At seed 42, 61.2% of its niches contained at least two annotated cell types, and 10 of the 11 lung-cancer cell types were represented. These cores therefore provide balanced examples for spatial maps and detailed compositional analyses while avoiding selection solely on niche heterogeneity or enrichment.

6. Discussion and Conclusion

HyperNiche provides a supervised framework for learning spatially localized, higher-order cellular neighborhoods without imposing cell-type homogeneity. Across the evaluated breast- and lung-cancer cohorts, the learned representations aligned with reference cell-type annotations, while the inferred hyperedges showed reproducibility across random seeds and captured spatially localized, compositionally selective neighborhoods. Together, these results establish that cell-type supervision can guide the learning of interpretable higher-order spatial structure and that the resulting niches can be characterized in terms of their membership stability, composition, enrichment, and compactness.

These findings should not be interpreted as independent biological validation of functional cellular niches. Cell-type annotations provided the supervision used to train HyperNiche and were also used in the subsequent compositional analyses. Moreover, the low normalized composition entropy indicates that many learned niches were dominated by one annotated cell type. The current results therefore demonstrate that the model can represent relationships spanning cell types, but do not establish that most inferred niches are heterophilic or functionally coordinated.

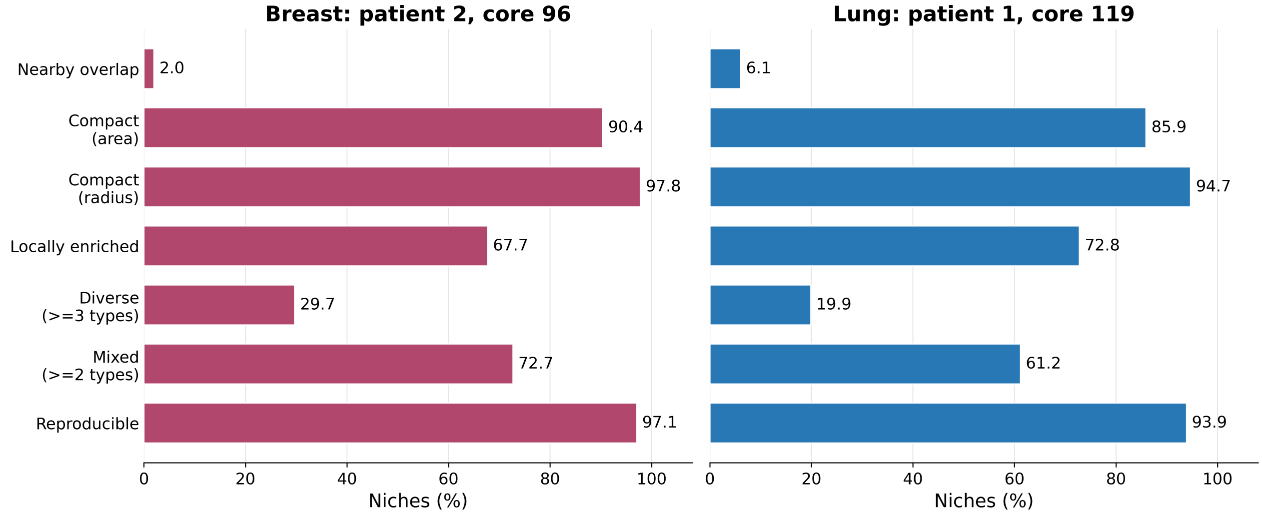

Fig. 2. Representative-core biological profiles for breast cancer (patient 2, core 96) and lung cancer (patient 1, core 119). Mixed and diverse niches contain at least two and at least three annotated cell types, respectively, in the seed-42 export. Nearby overlap is the mean membership overlap between spatially nearby niches, expressed as a percentage; lower values indicate less redundancy. Core selection considered the full set of reported properties and was not based on a single outcome.

Several additional limitations qualify the conclusions. Both cohorts contain a limited number of patients, restricting the assessment of population-level generalizability. The targeted Xenium gene panels provide spatially resolved measurements but capture only a subset of the transcriptome and may limit the molecular distinctions available to the model. Independent functional niche labels were unavailable, preventing direct evaluation of whether the inferred hyperedges correspond to experimentally established microenvironments. Under the stricter spatial null, enrichment was limited in the lung cohort and was not retained in the breast-cancer cohort, indicating that some apparent composition patterns may be explained by local tissue organization. Finally, constructing one candidate anchored hyperedge per cell may produce overlapping or partially redundant descriptions of the same local structure.

Future work should evaluate HyperNiche in larger, independent, and more diverse patient cohorts; incorporate broader transcriptomic or multimodal measurements; and test inferred niches against pathology, clinical outcomes, perturbation experiments, or independently defined functional microenvironments. Methods for merging, pruning, or selecting representative anchored hyperedges may also reduce redundancy and yield more concise tissue-level niche maps. Within these limitations, HyperNiche offers a flexible approach for learning and characterizing higher-order spatial cellular neighborhoods, while the present study provides evidence of structural consistency and biological plausibility rather than definitive functional validation.

Acknowledgments

The authors thank Dr. Kunal Rai and his team at MD Anderson Cancer Center for the many discussions that helped shape this research, and Dr. Hyeongseon Jeon and his team at the University of Houston for their feedback on the dataset.

References

1. C. G. Williams, H. J. Lee, T. Asatsuma, R. Vento-Tormo and A. Haque, An introduction to spatial transcriptomics for biomedical research, *Genome medicine* **14**, p. 68 (2022).
2. Y. Liao, C. Zhang, Z. Wang, F. Qi, W. Huang, S. Cai, J. Li, Z. Yuan, J. Song and H. Cai, Unveiling tissue structure and tumor microenvironment from spatially resolved transcriptomics by hypergraph learning, *bioRxiv* (2024).
3. K. Dong and S. Zhang, Deciphering spatial domains from spatially resolved transcriptomics with an adaptive graph attention auto-encoder, *Nature communications* **13**, p. 1739 (2022).
4. Y. Long, K. S. Ang, M. Li, K. L. K. Chong, R. Sethi, C. Zhong, H. Xu, Z. Ong, K. Sachaphibulkij, A. Chen, Z. Li, H. Fu, M. Wu, H. K. L. Lim, L. Liu and J. Chen, Deepst: A versatile graph contrastive learning framework for spatially informed clustering, integration, and deconvolution of spatial transcriptomics (2022).
5. M. Qu, Y. Zhao, G. Yang, D. Di, X. Su, H. Xu, Y. Song and L. Fan, Stag: Biologically guided spatial transcriptomics prediction via hypergraph learning, *Medical Image Analysis* , p. 104206 (2026).
6. J. Wang, X. Huang, H. Kuang and C. Yan, Subtype-hm: a novel cancer subtype identification method based on hypergraph learning and multi-omics data, *Interdisciplinary Sciences: Computational Life Sciences* , 1 (2025).
7. W. He, D. I. Bolnick, S. V. Scarpino and T. Eliassi-Rad, Hypergraph representations of single-cell rna sequencing data for improved cell clustering, *Bioinformatics* **42**, p. btag148 (2026).
8. Y. Kong and T. Yu, A hypergraph-based method for large-scale dynamic correlation study at the transcriptomic scale, *BMC genomics* **20**, p. 397 (2019).
9. E. Kulkul, T. Çukur and A. Koç, schyperlink: revealing cell-type-specific gene regulation with hypergraph neural networks, *IEEE Journal of Biomedical and Health Informatics* (2026).
10. B. Li, Y. Zhang, Q. Wang, C. Zhang, M. Li, G. Wang and Q. Song, Gene expression prediction from histology images via hypergraph neural networks, *Briefings in Bioinformatics* **25**, p. bbae500 (2024).
11. Z. Zhou, M. Zhou, X. Lin, X. Qi and G. Yan, Hypergraph neural diffusion: A pde-inspired framework for hypergraph message passing, *arXiv preprint arXiv:2604.10955* (2026).
12. B. Uncu and I. Yavuz, Scensemble: weighted hypergraph ensemble clustering for single-cell rna sequencing, *BMC bioinformatics* (2026).
13. L. Bai, Z. Li, C. Tang, C. Song and F. Hu, Hypergraph-based analysis of weighted gene co-expression hypernetwork, *Frontiers in Genetics* **16**, p. 1560841 (2025).
14. Y. Feng, H. You, Z. Zhang, R. Ji and Y. Gao, Hypergraph neural networks, *Proceedings of the AAAI Conference on Artificial Intelligence* **33**, 3558 (2019).
15. M. Soltani and L. Rueda, Hypergraph neural networks reveal spatial domains from single-cell transcriptomics data, *bioRxiv* , 2025 (2025).
16. C. Zhang, X. Li, B. Li, C. Deng, M. Li, S. Zhang, W. Yu, H. Zhang, Z. Wang, Y. Yang *et al.*, Hypergraph-driven spatial multimodal fusion for precise domain delineation and tumor microenvironment decoding, *Communications Biology* **9**, p. 45 (2025).
17. O. Eldaghar, Y. Zhu and D. Gleich, A spatial hypergraph model to smoothly interpolate between pairwise graphs and hypergraphs to study higher-order structures, *PLOS Complex Systems* **2**, p. e0000066 (2025).
18. X. Sun, C. Xu, J. F. Rocha, C. Liu, B. Hollander-Bodie, L. Goldman, M. DiStasio, M. Perlmutter and S. Krishnaswamy, Hyperedge representations with hypergraph wavelets: Applications to spatial transcriptomics, 1 (2025).
19. D. Zhou, J. Huang and B. Schölkopf, Learning with hypergraphs: Clustering, classification, and embedding, in *Advances in Neural Information Processing Systems*, (MIT Press, 2006).

20. H. Wang, R. Huang, J. Nelson, C. Gao, M. Tran, A. Yeaton, S. Krishna, K. Felt, K. L. Pfaff, T. Bowman, S. J. Rodig, K. Wei, B. A. Goods and S. L. Farhi, Systematic benchmarking of imaging spatial transcriptomics platforms in ffpe tissues, *Nature Communications* **16**, p. 10215 (2025).